

FEASIBILITY STUDY - NUMERIC OUTPUT

GLOBAL_PROBLEM_SIZE

grid_width: 5
grid_height: 5
grid_cells: 25
static_obstacles: 4
passable_cells: 21
scenarios: 8
scenario_probability_each: 0.1250000000000000
t_max: 10
mc_shots: 1000
qae_eval_qubits: 4
scenario_qubits: 3
target_qubits: 1
state_preparation_qubits: 4
qae_total_qubits: 8
qae_oracle_queries_per_sampler_shot: 15

PARAMETER_NOTES

qae_oracle_queries_per_sampler_shot: QAE checks the success oracle 15 times in one estimation round
exact_expected_score: exact average path quality across all scenarios; higher is better
qae_raw_estimation: direct QAE probability estimate from the discrete QAE grid
qae_mle: corrected QAE probability estimate; usually closer to the exact probability

GLOBAL_QUBIT_COUNTS

scenario_qubits: 3
target_qubits: 1
state_preparation_total_qubits: 4
qae_evaluation_qubits: 4
qae_total_qubits: 8
qae_classical_bits: 4

GATE_LEGEND

ccx: Toffoli gate / controlled-controlled-X gate
cp: controlled phase gate
cx: controlled-X gate / CNOT gate
h: Hadamard gate
mcphase: multi-controlled phase gate
measure: measurement operation
p: phase gate
s: S phase gate
sdg: inverse S phase gate
t: T phase gate
tdg: inverse T phase gate
u: general single-qubit unitary gate

PATH_REPORT: A*

PATH_STATS

path_cells: 9
path_moves: 8
unique_path_cells: 9
static_hits: 0
safe_scenarios: 5
unsafe_scenarios: 3
dynamic_collision_scenarios: 3
time_limit_fail_scenarios: 0
score_min: 0.0000000000000000
score_max: 0.2000000000000000
score_nonzero_count: 5
path_cells_list: [(0, 0), (0, 1), (0, 2), (0, 3), (0, 4), (1, 4), (2, 4), (3, 4), (4, 4)]

SCENARIO_NUMERIC_TABLE

scenario	dyn_count	delay	flight_time	dyn_hits	time_fail	safe	score
0	0	0	8	0	0	1	0.2000
1	1	0	8	0	0	1	0.2000
2	1	0	8	0	0	1	0.2000
3	1	1	9	1	0	0	0.0000
4	1	2	10	1	0	0	0.0000
5	2	0	8	0	0	1	0.2000
6	1	1	9	0	0	1	0.1000
7	1	0	8	1	0	0	0.0000

CLASSICAL_AND_QUANTUM_RESULTS

exact_safety_probability: 0.6250000000000000
exact_expected_score: 0.1125000000000000
monte_carlo_safety_probability: 0.6310000000000000
monte_carlo_expected_score: 0.1133000000000000

statevector_probability_target_1: 0.6249999999999984
qae_raw_estimation: 0.6913416999999999
qae_mle: 0.6232446871305140

METHOD_COMPARISON_PROBABILITY

method	probability	abs_error_vs_exact	calls_or_samples
exact_enumeration	0.6250000000000000	0.0000000000000000	8
monte_carlo_1000	0.6310000000000000	0.0060000000000000	1000
statevector_target	0.6249999999999984	0.0000000000000016	16
qae_raw_estimation	0.6913416999999999	0.0663416999999999	15360
qae_mle	0.6232446871305140	0.0017553128694860	15360

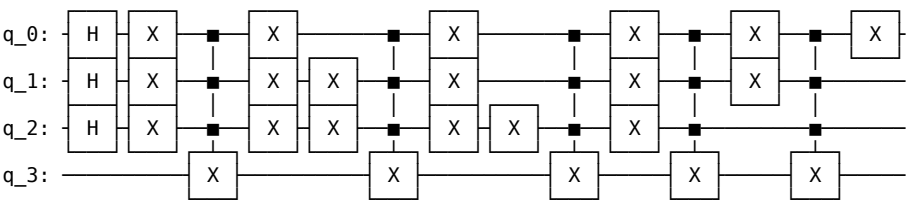
METHOD_ERROR_RANKING

rank_1: exact_enumeration, abs_error=0.0000000000000000
rank_2: statevector_target, abs_error=0.0000000000000016
rank_3: qae_mle, abs_error=0.0017553128694860
rank_4: monte_carlo_1000, abs_error=0.0060000000000000
rank_5: qae_raw_estimation, abs_error=0.0663416999999999

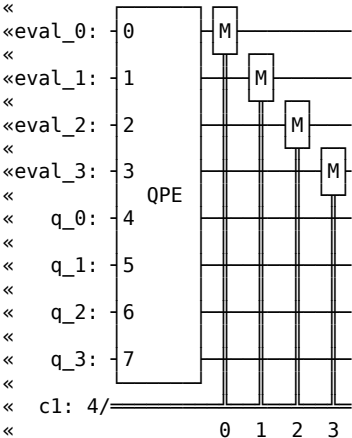
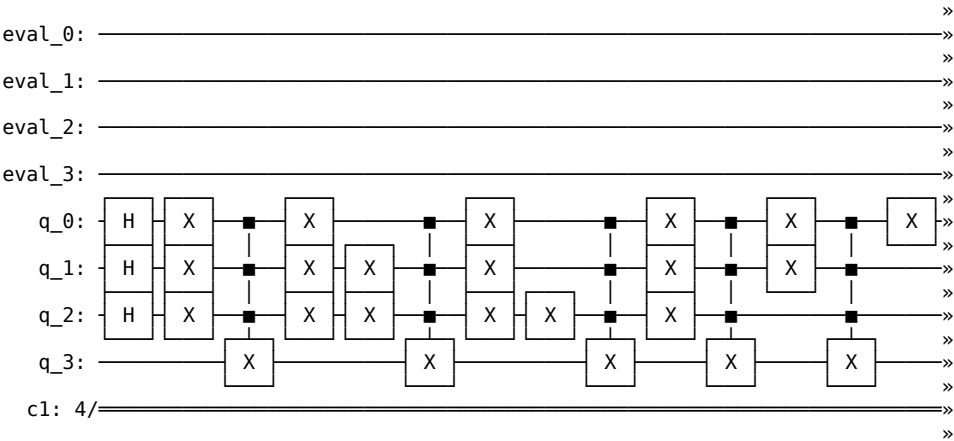
CIRCUIT_GATE_AND_DEPTH_SUMMARY

logical_A_depth: 14
logical_A_gate_count: 26
decomposed_A_depth: 131
decomposed_A_gate_count: 147
decomposed_A_gate_set: {cx, u}
logical_QAE_depth: 16
logical_QAE_gate_count: 31
decomposed_QAE_depth: 8445
decomposed_QAE_gate_count: 9015
decomposed_QAE_gate_set: {ccx, cp, cx, h, mcphase, measure, p, s, sdg, t, tdg, u}

STATE_PREPARATION_A_CIRCUIT



QAE_CIRCUIT



QAE_MEASUREMENTS

y | probability | cumulative probability

0	0.0029296875	0.0029296875
1	0.0048828125	0.0078125000
2	0.0048828125	0.0126953125
3	0.0126953125	0.0253906250
4	0.1025390625	0.1279296875
5	0.3222656250	0.4501953125
6	0.0292968750	0.4794921875
7	0.0078125000	0.4873046875
8	0.0078125000	0.4951171875
9	0.0097656250	0.5048828125
10	0.0166015625	0.5214843750
11	0.3203125000	0.8417968750
12	0.1230468750	0.9648437500
13	0.0224609375	0.9873046875
14	0.0078125000	0.9951171875
15	0.0048828125	1.0000000000

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PATH_REPORT: manual

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PATH_STATS

path_cells: 9
path_moves: 8
unique_path_cells: 9
static_hits: 0
safe_scenarios: 4
unsafe_scenarios: 4
dynamic_collision_scenarios: 4
time_limit_fail_scenarios: 0
score_min: 0.0000000000000000
score_max: 0.2000000000000000
score_nonzero_count: 3
path_cells_list: [(0, 0), (1, 0), (2, 0), (3, 0), (4, 0), (4, 1), (4, 2), (4, 3), (4, 4)]

SCENARIO_NUMERIC_TABLE

scenario	dyn_count	delay	flight_time	dyn_hits	time_fail	safe	score
0	0	0	8	0	0	1	0.2000
1	1	0	8	1	0	0	0.0000
2	1	0	8	1	0	0	0.0000
3	1	1	9	0	0	1	0.1000
4	1	2	10	0	0	1	0.0000
5	2	0	8	2	0	0	0.0000
6	1	1	9	1	0	0	0.0000
7	1	0	8	0	0	1	0.2000

CLASSICAL_AND_QUANTUM_RESULTS

exact_safety_probability: 0.5000000000000000
exact_expected_score: 0.0625000000000000
monte_carlo_safety_probability: 0.5090000000000000
monte_carlo_expected_score: 0.0589000000000000
statevector_probability_target_1: 0.4999999999999999
qae_raw_estimation: 0.5000000000000000
qae_mle: 0.5000000000000000

METHOD_COMPARISON_PROBABILITY

method	probability	abs_error_vs_exact	calls_or_samples
exact_enumeration	0.5000000000000000	0.0000000000000000	8
monte_carlo_1000	0.5090000000000000	0.0090000000000000	1000
statevector_target	0.4999999999999999	0.0000000000000011	16
qae_raw_estimation	0.5000000000000000	0.0000000000000000	15360
qae_mle	0.5000000000000000	0.0000000000000000	15360

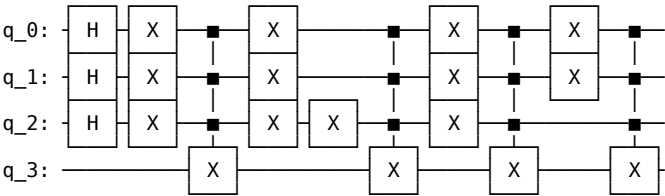
METHOD_ERROR_RANKING

rank_1: exact_enumeration, abs_error=0.0000000000000000
rank_2: qae_raw_estimation, abs_error=0.0000000000000000
rank_3: qae_mle, abs_error=0.0000000000000000
rank_4: statevector_target, abs_error=0.0000000000000011
rank_5: monte_carlo_1000, abs_error=0.0090000000000000

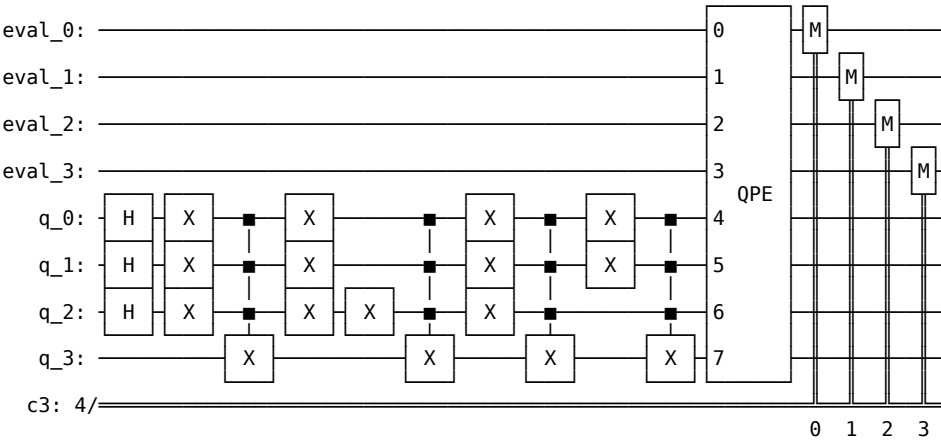
CIRCUIT_GATE_AND_DEPTH_SUMMARY

logical_A_depth: 10
logical_A_gate_count: 19
decomposed_A_depth: 105
decomposed_A_gate_count: 117
decomposed_A_gate_set: {cx, u}
logical_QAE_depth: 12
logical_QAE_gate_count: 24
decomposed_QAE_depth: 7038
decomposed_QAE_gate_count: 7508
decomposed_QAE_gate_set: {ccx, cp, cx, h, mcphase, measure, p, s, sdg, t, tdg, u}

STATE_PREPARATION_A_CIRCUIT



QAE_CIRCUIT



QAE_MEASUREMENTS

y	probability	cumulative probability
4	0.5048828125	0.5048828125
12	0.4951171875	1.0000000000

FINAL_COMPARISON

COMPARISON_NOTE

path_quality_metrics: directly comparable because both paths use the same scenarios and parameters
circuit_resource_metrics: path-dependent because each path creates a different oracle/state-preparation circuit

PATH_QUALITY_COMPARISON

metric	A*	manual	better
exact_safety_probability	0.6250000000000000	0.5000000000000000	A*
exact_expected_score	0.1125000000000000	0.0625000000000000	A*
monte_carlo_safety_probability	0.6310000000000000	0.5090000000000000	A*
monte_carlo_expected_score	0.1133000000000000	0.0589000000000000	A*
qae_mle	0.6232446871305140	0.5000000000000000	A*
safe_scenarios	5	4	A*
unsafe_scenarios	3	4	A*
dynamic_collision_scenarios	3	4	A*
score_nonzero_count	5	3	A*